Assignment No - 01

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Section : 10

Course : CSE260

Ans to the Q No-1

(a)

$$(101110010001)_2 = (1\times2^0 + 1\times2^4 + 1\times2^7 + 1\times2^8 + 1\times2^9 + 1\times2^{11})_{10}$$

$$= (2961)_{10}$$

(b)

$$(11011.101)_2 = (1\times2^0 + 1\times2^1 + 1\times2^3 + 1\times2^4) + (1\times2^{-1} + 1\times2^{-3})$$

$$= (27 + 0.625)_{10}$$

$$= (27.625)_{10}$$

(Ans)

Ans to the Q No-2

$(4195)_{10}$,

2	4195	
2	2097	1
2	1048	1
2	524	0
2	262	0
2	131	0
2	65	1
2	32	1
2	16	0
2	8	0
2	4	0
2	2	0
2	1	0
	0	1

So,

$$(4195)_{10} = (1000001100011)_2.$$

(Ans)

Ans to the Q No-3

$$\begin{array}{r|l} 16 & 513 \\ \hline 16 & 32 - 1 \\ \hline 16 & 2 - 0 \\ \hline & 0 - 2 \end{array}$$

$\therefore (513)_{10} = (201)_{16}$.

(Ans)

Ans to the Q No-4

(a) $(29)_{12} = 9\times12^0 + 2\times12^1 = (33)_{10}$

Now,

$$\begin{array}{r|l} 7 & 33 \\ \hline 7 & 4 - 5 \\ \hline & 0 - 4 \end{array}$$

$\therefore (29)_{12} = (45)_7$.

(b) $(10110111)_5 = 5^0 + 5^1 + 5^2 + 5^4 + 5^5 + 5^7 = (81906)_{10}$.

$$\begin{array}{r|l} 4 & 81906 \\ \hline 4 & 20476 - 2 \\ \hline 4 & 5119 - 0 \\ \hline 4 & 1279 - 3 \\ \hline 4 & 319 - 3 \\ \hline 4 & 79 - 3 \\ \hline 4 & 19 - 3 \\ \hline 4 & 4 - 3 \\ \hline 4 & 1 - 0 \\ \hline & 0 - 1 \end{array}$$

$\therefore (10110111)_5 = (103333302)_4$. (Ans)

Ans to the Q No-5

Addition:

$$
\begin{array}{r}
(412)_9 \\
+\ (134)_9 \\
\hline
(546)_9
\end{array}
$$

Now, $(412)_9 = 2\times9^0 + 1\times9^1 + 4\times9^2 = (335)_{10}$

$(134)_9 = 4\times9^0 + 3\times9^1 + 1\times9^2 = (112)_{10}$

$(546)_9 = 6\times9^0 + 4\times9^1 + 5\times9^2 = (447)_{10}$

So, $(335)_{10} + (112)_{10} = (447)_{10}$

Addition is verified.

Subtraction:

$$
\begin{array}{r}
(412)_9 \\
-\ (134)_9 \\
\hline
(267)_9
\end{array}
$$

$(267)_9 = 7\times9^0 + 6\times9^1 + 2\times9^2 = (223)_{10}$

Now, $(335)_{10} - (112)_{10} = (223)_{10}$

$\therefore$ Subtraction is verified.

Multiplication

$$\begin{array}{r} (412)_9 \\ \times\ (134)_9 \\ \hline 1748 \\ 1336- \\ 412-- \\ \hline (56418)_9 \end{array}$$

$(56418)_9 = 8\times9^0 + 1\times9^1 + 4\times9^2 + 6\times9^3 + 5\times9^4 = (37520)_{10}$.

Now, $(335)_{10} \times (112)_{10} = (37520)_{10}$.

Multiplication is verified.

Ans

Ans to the Q No-6

$(27)_{10} = (11011)_2$

$+27 = 0011011$ (7 bits)

$(13)_{10} = (1101)_2$

$+13 = 000\ 1101$ (7 bits)

↓ 1's compliment

$111\ 0010$

$+\ 1$

$(-13) = 111\ 0011$ (2's compliment)

Now, $27 + (-13) =$

$$\begin{array}{r} 0011011 \\ 1110011 \\ \hline \underbrace{1}_{\text{carry}}0001110 \end{array}$$

Without carry result is 0001110.

Here we are subtracting 13 from 27 so the answer should be positive. The result we got is positive. So no overflow.

Ans

Ans to the Q No-7

$(13)_{10} = (1101)_2$ $\qquad$ $(27)_{10} = 11011$

$+13 = 001101$ (6 bits) $\qquad$ $+27 = 011011$, (6 bits)

Now

$$
\begin{array}{r}
13+27 = 001101 \\
011011 \\
\hline
101000
\end{array}
$$

$\therefore$ The result is 101000

Here we are adding two positive numbers but the result is negative

So there is an overflow.

(Ans)

Ans to the Q No-8

(a)

Using 2's compliment

$91 = 1011011$

$+91 = 000\ 0000\ 0101\ 1011$ (15 bits)

$499 = 1\ 1111\ 0011$

$+499 = 000\ 0001\ 1111\ 0011$ (15 bits)

$\big\downarrow$ 1's compliment

$$
\begin{array}{r}
111\ 1110\ 0000\ 1100 \\
+\ 1 \\
\hline
\end{array}
$$

$(-499) = 111\ 1110\ 0000\ 1101$ (2's compliment)

$$
\begin{array}{lcccc}
91+(-499) = & 000 & 0000 & 0101 & 1011 \\
 & 111 & 1110 & 0000 & 1101 \\
\hline
 & 111 & 1110 & 0110 & 1000
\end{array}
$$

If we subtract 499 from 91 the answer should be negative. Here the result is negative. So no overflow.

Using 1's compliment

$+91 = 000\ 0000\ 0101\ 1011$

$+499 = 000\ 0001\ 1111\ 0011$

$\downarrow$ 1's compliment

$-499 = 111\ 1110\ 0000\ 1100$

$$
\begin{array}{lcccc}
91+(-499) = & 000 & 0000 & 0101 & 1011 \\
 & 111 & 1110 & 0000 & 1100 \\
\hline
 & 111 & 1110 & 0110 & 0111
\end{array}
$$

Here the result is also negative. So there is no overflow.

(b) Using 2's compliment

$379 = 1\ 0111\ 1011$

$+379 = 000\ 0001\ 0111\ 1011$ (15 bits)

$98 = 110\ 0010$

$+98 = 000\ 0000\ 0110\ 0010$ (15 bits)

$$
\begin{array}{lcccc}
379 + 98 = & 000 & 0001 & 0111 & 1011 \\
 & 000 & 0000 & 0110 & 0010 \\
\hline
 & 000 & 0001 & 1101 & 1101
\end{array}
$$

We are adding two positive number the answer should be positive

Here the result is positive so no overflow.

<u>Using 1's compliments</u>

$$
\begin{array}{lcccc}
379 + 98 = & 000 & 0001 & 0111 & 1011 \\
 & 000 & 0000 & 0110 & 0010 \\
\hline
 & 000 & 0001 & 1101 & 1101
\end{array}
$$

As it is the same - So no overflow occurs.

(Ans)

Ans to the Q No-9

Each RAM cost $= (1C2)_{16} = (2\times16^0 + 12\times16^1 + 1\times16^2)_{10}$ dollars

$= (450)_{10}$ dollars.

Two RAM will cost $= 2\times450 = 900$ dollars.

Graphics card cost $= (10010110000)_2 = (2^4+2^5+2^7+2^{10})_{10}$ dollars

$= (1200)_{10}$ dollars.

Total cost $= (1200+900)$ dollars

$= (2100)_{10}$ dollars

Lent money from friend $= (4064)_8 = ((4\times8^0)+(6\times8^1)+(4\times8^3))_{10}$ dollars

$= (2100)_{10}$ dollars.

Money will be left $= (2100-2100)$ dollars

$= (0)_{10}$ dollars.

So after buying those components no money will be left.

Ans.